



Assessment Submission Form

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Signed.....Aditya..... Date18.09.2026.....

Table of Contents

Introduction.....	2
System Design	4
Implementation	6
Results.....	7
Reflection.....	15
Appendix.....	16

Introduction

The project takes up a gym as the object of interest and puts forward a system to manage certain entities including members, trainers, workout plans and payments. Manual handling

of these data might complicate their arrangement and result in mistakes or even loss of data.

The goal of this project is to create a simple Gym Management System with Python providing a way of managing these important aspects.

This software application works in command line mode and has a menu where one can choose different actions. The functionality includes the ability to register a new member, view and search among member's records, edit member's contact details, add a trainer, assign a trainer to a member, create workout plans, add exercises and make payments. Also, some basic validation of input is done by the program to avoid invalid actions.

The scope is limited to only the essential administration actions in a small gym. The information about members, trainers, workout plans and payments is stored in .csv files providing persistence between the launches of the program. This project demonstrates only basic Python programming and object-oriented programming skills.

System Design

This system consists of separate Python classes each one has its own responsibility. The main classes include :-

1. Member
2. Trainer
3. Workoutplan
4. Payment
5. GymManager

The trainer class describes a gym trainer. This class contains a trainer ID and name as well as the list of assigned members. There are methods such as `assign_client`, `remove_client` and `get_client_count`.

The workoutplan class helps to control workout plans. A workout plan consists of a plan ID, plan name, its difficulty and the list of exercises. Methods which allow adding, removing exercises and changing plan difficulty.

The payment class describes a payment transaction. It contains a payment ID, member ID, payment amount and the status. Methods to change payment status and payment amount.

The GymManager class manages data used in the whole system. It creates file paths for four CSV files and stores loaded objects in dictionaries. Loaded data are read at the beginning of the program execution and saved back to CSV files.

Short overview of relationships between classes is given below:

The interactive.py file provides user interface and connects all mentioned classes via menu functions which are responsible for member management, trainer management, workout plans and payment management.

Implementation

The application design is divided into different Python modules that are made as per to perform specific tasks.

The member management module consists of different actions including registration, view, search and edit operations. The `search_member()` function searches for stored member objects either by id or name.

In the Gym Management System the `GymManager` class uses different dictionaries for storing objects; there are different dictionaries for members, trainers, workout plans and payments which allows for quick access to records by means of their ids.

List is used when multiple values need to be stored. Specifically, the trainer class uses list for storing assigned clients and the workout plan class uses list for storing exercise records. File management is an integral part of this project and CSV files are used for storing data. When the application starts, the `load_all_data()` function loads the records and when information is added or edited, `save_all_data()` writes the current data to the CSV files.

There is exception handling that helps deal with such problems as invalid data and file errors. In particular, the process of data loading includes handling of `IOError`, `KeyError`, and `ValueError`. Also, the payment management module includes handling of invalid monetary values input by users.

`Interactive.py` file includes the interface and the separate modules include different classes and data management functions. Use of clear variable and function names along with comments and a menu make the code more understandable.

Results

1. To display all the members

```
GYM MANAGEMENT SYSTEM (CLI)
1. Member Management
2. Trainer Management
3. Workout Plans
4. Payment Management
5. Exit
Select a section (1-5): 1

--- Member Management ---
1. Register member
2. View all members
3. Search member
4. Update member Contact
5. Return to main menu
Select an option (1-5): 2
ID: M001 | Name: John Smith | Email: john.smith@gmail.com | Phone: +491234567890 | Trainer: none
ID: M002 | Name: Will Smith | Email: will.smith@gmail.com | Phone: +351762096762 | Trainer: none
ID: M003 | Name: Tom Hardy | Email: tomi@gmail.com | Phone: +491745556789 | Trainer: T01
ID: M004 | Name: Jimmy Sandhu | Email: sandhu34@gmail.com | Phone: +491234567890 | Trainer: none
ID: M005 | Name: Ammy James | Email: Ammy12@gmail.com | Phone: +491234567890 | Trainer: none
```

2. Search Member based on Name and ID

```
--- Member Management ---
1. Register member
2. View all members
3. Search member
4. Update member Contact
5. Return to main menu
Select an option (1-5): 3
Enter member ID or Name to search: Will
Found: ID: M002 | Name: Will Smith | Trainer: none
```

```
--- Member Management ---
1. Register member
2. View all members
3. Search member
4. Update member Contact
5. Return to main menu
Select an option (1-5): 3
Enter member ID or Name to search: M004
Found: ID: M004 | Name: Jimmy Sandhu | Trainer: none
```

3. To update the contact – used blank to make no change in email.

```
--- Member Management ---
1. Register member
2. View all members
3. Search member
4. Update member Contact
5. Return to main menu
Select an option (1-5): 4
Enter Member ID: M005
Enter new email (leave blank to skip): Mummyammy@gmail.com
Enter new phone (leave blank to skip):
Member details updated

--- Member Management ---
1. Register member
2. View all members
3. Search member
4. Update member Contact
5. Return to main menu
Select an option (1-5): 2
ID: M001 | Name: John Smith | Email: john.smith@gmail.com | Phone: +491234567890 | Trainer: none
ID: M002 | Name: Will Smith | Email: will.smith@gmail.com | Phone: +351762096762 | Trainer: none
ID: M003 | Name: Tom Hardy | Email: tomi@gmail.com | Phone: +491745556789 | Trainer: T01
ID: M004 | Name: Jimmy Sandhu | Email: sandhu34@gmail.com | Phone: +491234567890 | Trainer: none
ID: M005 | Name: Ammy James | Email: Mummyammy@gmail.com | Phone: +491234567890 | Trainer: none
```

4. Register a new member

```
GYM MANAGEMENT SYSTEM (CLI)
1. Member Management
2. Trainer Management
3. Workout Plans
4. Payment Management
5. Exit
Select a section (1-5): 1

--- Member Management ---
1. Register member
2. View all members
3. Search member
4. Update member Contact
5. Return to main menu
Select an option (1-5): 1
Enter member id: M006
Enter name: Jaslin
Enter email: jai@gmail.com
Enter phone: +4917645678903
Member: 'Jaslin' registered

--- Member Management ---
1. Register member
2. View all members
3. Search member
4. Update member Contact
5. Return to main menu
Select an option (1-5): 2
ID: M001 | Name: John Smith | Email: john.smith@gmail.com | Phone: +491234567890 | Trainer: none
ID: M002 | Name: Will Smith | Email: will.smith@gmail.com | Phone: +351762096762 | Trainer: none
ID: M003 | Name: Tom Hardy | Email: tomi@gmail.com | Phone: +491745556789 | Trainer: T01
ID: M004 | Name: Jimmy Sandhu | Email: sandhu34@gmail.com | Phone: +491234567890 | Trainer: none
ID: M005 | Name: Ammy James | Email: Mummyammy@gmail.com | Phone: +491234567890 | Trainer: none
ID: M006 | Name: Jaslin | Email: jai@gmail.com | Phone: +4917645678903 | Trainer: none
```

5. Register and view all trainers

```
GYM MANAGEMENT SYSTEM (CLI)
1. Member Management
2. Trainer Management
3. Workout Plans
4. Payment Management
5. Exit
Select a section (1-5): 2

--- Trainer Management ---
1. Add Trainer
2. View Trainer
3. Assign Trainer to Member
4. Return to Main Menu
Select an option (1-4): 1
Enter Trainer ID: T04
Enter Name: White Devil
Trainer 'White Devil' added

--- Trainer Management ---
1. Add Trainer
2. View Trainer
3. Assign Trainer to Member
4. Return to Main Menu
Select an option (1-4): 2
ID: T01 | Name: Guri Sandhu | Total Clients: 1
ID: T02 | Name: Harry Singh | Total Clients: 0
ID: T03 | Name: Black Devil | Total Clients: 0
ID: T04 | Name: White Devil | Total Clients: 0
```

6. Assign a new trainer

```
--- Trainer Management ---
1. Add Trainer
2. View Trainer
3. Assign Trainer to Member
4. Return to Main Menu
Select an option (1-4): 3
Enter Member ID: M003
Enter Trainer ID: T02
Trainer assigned successfully
```

7. Create and view plans

```
GYM MANAGEMENT SYSTEM (CLI)
1. Member Management
2. Trainer Management
3. Workout Plans
4. Payment Management
5. Exit
Select a section (1-5): 3

--- Workout Plans ---
1. Create Plan
2. Add Exercise to Plan
3. View Plans
4. Return to Main Menu
Select an option (1-4): 1
Enter plan ID: P05
Enter plan name: Push Pull Legs
Difficulty (Beginner/Intermediate/Advanced): Intermediate
Workout plan created

--- Workout Plans ---
1. Create Plan
2. Add Exercise to Plan
3. View Plans
4. Return to Main Menu
Select an option (1-4): 3
ID: P01 | Name: Strength Basics | Difficulty: Beginner | Exercises: None
ID: P02 | Name: Gymnastics Max | Difficulty: Advanced | Exercises: None
ID: P03 | Name: Cardio Burn | Difficulty: Intermediate | Exercises: None
ID: P04 | Name: Mobility Flow | Difficulty: Beginner | Exercises: None
ID: P05 | Name: Push Pull Legs | Difficulty: Intermediate | Exercises: None
```

8. Add exercise to plan

```
--- Workout Plans ---  
1. Create Plan  
2. Add Exercise to Plan  
3. View Plans  
4. Return to Main Menu  
Select an option (1-4): 2  
Enter Plan ID: P05  
Enter Exercise Name: Legs  
Added 'Legs' to plan
```

```
--- Workout Plans ---  
1. Create Plan  
2. Add Exercise to Plan  
3. View Plans  
4. Return to Main Menu  
Select an option (1-4): 3  
ID: P01 | Name: Strength Basics | Difficulty: Beginner | Exercises: None  
ID: P02 | Name: Gymnastics Max | Difficulty: Advanced | Exercises: None  
ID: P03 | Name: Cardio Burn | Difficulty: Intermediate | Exercises: None  
ID: P04 | Name: Mobility Flow | Difficulty: Beginner | Exercises: None  
ID: P05 | Name: Push Pull Legs | Difficulty: Intermediate | Exercises: Legs
```

9. Recording a payment

```
--- Payment & Billing ---
1. Make/Record Payment
2. View All Payments
3. Check Payment Status
4. Return to Main Menu
Select an option (1-4): 2
ID: PAY101 | Member: M001 | Amount: $50.00 | Status: Paid
ID: PAY102 | Member: M002 | Amount: $50.00 | Status: Paid
ID: PAY103 | Member: M003 | Amount: $75.00 | Status: Pending
ID: PAY104 | Member: M004 | Amount: $50.00 | Status: Failed
ID: PAY105 | Member: M005 | Amount: $150.00 | Status: Paid

--- Payment & Billing ---
1. Make/Record Payment
2. View All Payments
3. Check Payment Status
4. Return to Main Menu
Select an option (1-4): 1
Enter Payment ID: PAY106
Enter Member ID: M006
Enter Amount: 75
Status (Paid/Pending/Failed) [Default: Paid]: Paid
Payment recorded

--- Payment & Billing ---
1. Make/Record Payment
2. View All Payments
3. Check Payment Status
4. Return to Main Menu
Select an option (1-4): 2
ID: PAY101 | Member: M001 | Amount: $50.00 | Status: Paid
ID: PAY102 | Member: M002 | Amount: $50.00 | Status: Paid
ID: PAY103 | Member: M003 | Amount: $75.00 | Status: Pending
ID: PAY104 | Member: M004 | Amount: $50.00 | Status: Failed
ID: PAY105 | Member: M005 | Amount: $150.00 | Status: Paid
ID: PAY106 | Member: M006 | Amount: $75.00 | Status: Paid
```

10. Payment Status enquiry

```
--- Payment & Billing ---
1. Make/Record Payment
2. View All Payments
3. Check Payment Status
4. Return to Main Menu
Select an option (1-4): 3
Enter Payment ID: PAY105
Payment PAY105 for Member M005: $150.00 [Paid]

--- Payment & Billing ---
1. Make/Record Payment
2. View All Payments
3. Check Payment Status
4. Return to Main Menu
Select an option (1-4): 4
```

11. Exiting

```
=====
GYM MANAGEMENT SYSTEM (CLI)
1. Member Management
2. Trainer Management
3. Workout Plans
4. Payment Management
5. Exit
Select a section (1-5): 5
System state saved. Exiting.
```

12. Error Handling

```
--- Member Management ---
1. Register member
2. View all members
3. Search member
4. Update member Contact
5. Return to main menu
Select an option (1-5): 6
Choose 1-5 !!!

--- Member Management ---
1. Register member
2. View all members
3. Search member
4. Update member Contact
5. Return to main menu
Select an option (1-5): 1
Enter member id: M005
Member id already exists
```

13. No Member Found Error

```
--- Member Management ---
1. Register member
2. View all members
3. Search member
4. Update member Contact
5. Return to main menu
Select an option (1-5): 3
Enter member ID or Name to search: M009
No matching member found
```

14. Payment not found

```
GYM MANAGEMENT SYSTEM (CLI)
1. Member Management
2. Trainer Management
3. Workout Plans
4. Payment Management
5. Exit
Select a section (1-5): 4

--- Payment & Billing ---
1. Make/Record Payment
2. View All Payments
3. Check Payment Status
4. Return to Main Menu
Select an option (1-4): 3
Enter Payment ID: 4
Payment record not found
```

15. Invalid ID

```
--- Trainer Management ---
1. Add Trainer
2. View Trainer
3. Assign Trainer to Member
4. Return to Main Menu
Select an option (1-4): 3
Enter Member ID: M009
Enter Trainer ID: T02
Assignment failed (invalid Member ID or Trainer ID)
```

Reflection

The project helped improve knowledge about programming in Python language especially concerning classes, functions, dictionaries, and handling of CSV files. One of the major challenges was making sure that the program could successfully read from and write into the data files. The project has also shown that splitting the code in several files helps make it manageable.

In the future, one can use a database in place of CSV and implement other features like keeping track of attendances and expiry date of membership.

Appendix

Github Link :-

<https://github.com/avengeradi/Gisma/tree/9e351315b916f7c7cc5649601092d8983ed5a842/B100A%20Python%20Programming%20WS0926>